

DSC5001: Statistical Machine Learning I

Assignment 1 – Solutions

Semester A, 2026/27

Total: 100 marks

Question	1	2	3	4	5	6	7	8	Total
Marks	10	10	10	20	10	10	10	20	100

Instructor solution set

Solution to Question 1: Gaussian Geometry and Linear Combinations

The covariance matrix is

$$\Sigma = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}, \quad \det(\Sigma) = 16 - 9 = 7,$$

so

$$\Sigma^{-1} = \frac{1}{7} \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix}.$$

(a) Because the mean is zero, the squared Mahalanobis distance is

$$d^2(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \mathbf{x}.$$

For $\mathbf{x}_A = (2, 2)^\top$,

$$\Sigma^{-1} \mathbf{x}_A = \frac{1}{7} \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/7 \\ 2/7 \end{bmatrix},$$

and hence

$$d_A^2 = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} 2/7 \\ 2/7 \end{bmatrix} = \boxed{\frac{8}{7}}.$$

For $\mathbf{x}_B = (2, -2)^\top$,

$$\Sigma^{-1} \mathbf{x}_B = \frac{1}{7} \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix},$$

so

$$d_B^2 = \begin{bmatrix} 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \boxed{8}.$$

(b) For two points under the same multivariate Gaussian, the normalizing constant is identical, so the density ratio depends only on the Mahalanobis terms:

$$\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)} = \exp \left[-\frac{1}{2} (d_A^2 - d_B^2) \right].$$

Therefore,

$$\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)} = \exp \left[-\frac{1}{2} \left(\frac{8}{7} - 8 \right) \right] = \boxed{\exp \left(\frac{24}{7} \right)} \approx 30.83.$$

Thus $\boxed{\mathbf{x}_A}$ has the larger Gaussian density.

Notice that $\|\mathbf{x}_A\|_2 = \|\mathbf{x}_B\|_2$, so Euclidean distance alone cannot distinguish them.

(c) Using

$$\text{Var}(aX_1 + bX_2) = a^2 \text{Var}(X_1) + b^2 \text{Var}(X_2) + 2ab \text{Cov}(X_1, X_2),$$

we obtain

$$\text{Var}(U) = 4 + 4 + 2(3) = \boxed{14},$$

and

$$\text{Var}(V) = 4 + 4 - 2(3) = \boxed{2}.$$

Also,

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X_1 + X_2, X_1 - X_2) \\ &= \text{Var}(X_1) - \text{Cov}(X_1, X_2) + \text{Cov}(X_2, X_1) - \text{Var}(X_2) \\ &= 4 - 3 + 3 - 4 \\ &= \boxed{0}. \end{aligned}$$

Since (U, V) is a linear transformation of a jointly Gaussian vector, it is jointly Gaussian. For jointly Gaussian variables, zero covariance implies independence. Hence

U and V are independent.

Solution to Question 2: Method of Moments versus Maximum Likelihood

The model is

$$f(x | \theta) = \frac{2x}{\theta^2}, \quad 0 \leq x \leq \theta.$$

(a) The first population moment is

$$\mathbb{E}_\theta[X] = \int_0^\theta x \frac{2x}{\theta^2} dx = \frac{2}{\theta^2} \int_0^\theta x^2 dx = \frac{2\theta}{3}.$$

The sample mean is

$$\bar{x} = \frac{1 + 2 + 3 + 4}{4} = \frac{5}{2}.$$

Method of Moments sets

$$\frac{2\theta}{3} = \bar{x},$$

so

$$\hat{\theta}_{\text{MM}} = \frac{3}{2}\bar{x} = \boxed{\frac{15}{4} = 3.75}.$$

(b) For a sample $x_1, \dots, x_n$,

$$L(\theta) = \prod_{i=1}^n \frac{2x_i}{\theta^2} \mathbf{1}\{\theta \geq x_i\}.$$

For the observed sample,

$$x_{(n)} = \max_i x_i = 4,$$

so

$$L(\theta) = \frac{2^4(1)(2)(3)(4)}{\theta^8} \mathbf{1}\{\theta \geq 4\} = \frac{384}{\theta^8} \mathbf{1}\{\theta \geq 4\}.$$

On the admissible region $\theta \geq 4$, the function θ^{-8} is strictly decreasing. Therefore the likelihood is maximized at the smallest admissible value:

$$\boxed{\hat{\theta}_{\text{MLE}} = 4}.$$

(c) The two estimates are

$$\hat{\theta}_{\text{MM}} = 3.75, \quad \hat{\theta}_{\text{MLE}} = 4.$$

The MoM estimate gives

$$\hat{\theta}_{\text{MM}} < \max_i x_i = 4.$$

Under the model, observations must satisfy $X \leq \theta$. Hence the observation $x = 4$ is impossible when $\theta = 3.75$, and

$$\boxed{L(3.75) = 0}.$$

The MLE explicitly respects the support of the likelihood and chooses the boundary value $\theta = 4$.

This example shows that moment matching uses selected distributional summaries but does not automatically enforce every structural constraint implied by the full probability model.

Solution to Question 3: Bootstrap Uncertainty for a Range Statistic

For the observed sample

$$D = \{1, 3, 4, 8, 10\},$$

the observed range is

$$\hat{\theta} = T(D) = 10 - 1 = \boxed{9}.$$

(a) The eight bootstrap ranges are

b	$D^{*(b)}$	$\hat{\theta}^{*(b)}$
1	(1, 3, 3, 4, 8)	7
2	(1, 1, 4, 10, 10)	9
3	(3, 3, 4, 8, 8)	5
4	(1, 4, 4, 8, 10)	9
5	(1, 1, 3, 4, 4)	3
6	(3, 4, 8, 8, 10)	7
7	(1, 3, 8, 10, 10)	9
8	(1, 3, 4, 4, 8)	7

Thus

$$\boxed{\hat{\theta}^* = (7, 9, 5, 9, 3, 7, 9, 7)}.$$

(b) Their mean is

$$\overline{\hat{\theta}^*} = \frac{7 + 9 + 5 + 9 + 3 + 7 + 9 + 7}{8} = \boxed{7}.$$

The bootstrap bias estimate is

$$\widehat{\text{Bias}}_{\text{boot}} = \overline{\hat{\theta}^*} - \hat{\theta} = 7 - 9 = \boxed{-2}.$$

For the bootstrap standard error,

$$\sum_{b=1}^8 \left(\hat{\theta}^{*(b)} - 7 \right)^2 = 0^2 + 2^2 + (-2)^2 + 2^2 + (-4)^2 + 0^2 + 2^2 + 0^2 = 32.$$

Therefore,

$$\widehat{\text{SE}}_{\text{boot}} = \sqrt{\frac{32}{8-1}} = \sqrt{\frac{32}{7}} \approx \boxed{2.138}.$$

(c) If five observations are sampled *without replacement* from a dataset containing exactly five observations, every resample contains all five original values, only possibly in a different order.

Therefore every resample has range

$$10 - 1 = 9,$$

so the estimated standard error would be

$$\boxed{0}.$$

This does not reproduce the sampling variation that the ordinary nonparametric bootstrap is designed to mimic. Ordinary bootstrap sampling is *with replacement*; repetitions and omissions are what allow the empirical distribution to generate different pseudo-samples.

Solution to Question 4: Regression Fitting, Leverage, and Influence

The data are

$$x = (0, 1, 2, 3, 4, 9), \quad y = (1, 3, 8, 7, 9, 18), \quad n = 6.$$

(a) First,

$$\bar{x} = \frac{19}{6}, \quad \bar{y} = \frac{46}{6} = \frac{23}{3}.$$

The centered sums are

$$S_{xx} = \sum_i (x_i - \bar{x})^2 = \frac{305}{6},$$

and

$$S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \frac{277}{3}.$$

Hence

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{554}{305} \approx 1.8164,$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{584}{305} \approx 1.9148.$$

Therefore the fitted line is

$$\boxed{\hat{y} = 1.9148 + 1.8164x}.$$

(b) The fitted values and residuals are

i	x_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$
1	0	1.9148	-0.9148
2	1	3.7311	-0.7311
3	2	5.5475	2.4525
4	3	7.3639	-0.3639
5	4	9.1803	-0.1803
6	9	18.2623	-0.2623

More exactly,

$$RSS = \sum_i e_i^2 = \frac{2324}{305} \approx \boxed{7.6197}.$$

Since two regression coefficients are estimated,

$$\hat{\sigma}^2 = \frac{RSS}{n-2} = \frac{2324/305}{4} = \frac{581}{305} \approx \boxed{1.9049}.$$

(c) For simple regression with an intercept,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

The six leverage values are

$$\boxed{(h_{11}, \dots, h_{66}) \approx (0.3639, 0.2590, 0.1934, 0.1672, 0.1803, 0.8361)}.$$

As a check,

$$\sum_{i=1}^6 h_{ii} = 2,$$

which equals the number of fitted regression coefficients.

The largest leverage is

$$h_{66} \approx 0.8361,$$

corresponding to $(x_6, y_6) = (9, 18)$.

(d) Using the proportional Cook's-distance score

$$C_i = e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2},$$

we obtain

$$(C_1, \dots, C_6) \approx (0.7527, 0.2522, 1.7885, 0.0319, 0.0087, 2.1403).$$

Thus the most influential observation is

$$i = 6.$$

However, the largest absolute residual is

$$|e_3| \approx 2.4525,$$

whereas

$$|e_6| \approx 0.2623.$$

Therefore the most influential point is *not* the point with the largest residual. Observation 6 becomes highly influential because its leverage is extremely large.

(e) Removing observation 6 leaves

$$x = (0, 1, 2, 3, 4), \quad y = (1, 3, 8, 7, 9).$$

For these five observations,

$$\bar{x} = 2, \quad \bar{y} = 5.6, \quad S_{xx} = 10, \quad S_{xy} = 20.$$

Hence

$$\hat{\beta}_1^{(-6)} = \frac{20}{10} = 2,$$

and

$$\hat{\beta}_0^{(-6)} = 5.6 - 2(2) = 1.6.$$

The refitted line is

$$\hat{y}^{(-6)} = 1.6 + 2x.$$

The slope changes from approximately

$$1.8164 \quad \text{to} \quad 2.0000.$$

Thus observation 6 has a relatively small residual but still changes the fitted relationship noticeably because of its extreme input location. This is precisely the distinction between residual size and influence.

Solution to Question 5: Mean and Covariance under a Linear Transformation

Let

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}.$$

(a) By linearity of expectation,

$$\begin{aligned}\mathbb{E}[\mathbf{Y}] &= \mathbb{E}[\mathbf{A}\mathbf{X} + \mathbf{b}] \\ &= \mathbf{A}\mathbb{E}[\mathbf{X}] + \mathbf{b} \\ &= \boxed{\mathbf{A}\boldsymbol{\mu} + \mathbf{b}}.\end{aligned}$$

(b) Let

$$\boldsymbol{\mu}_Y = \mathbb{E}[\mathbf{Y}] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}.$$

Then

$$\begin{aligned}\text{Cov}(\mathbf{Y}) &= \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top] \\ &= \mathbb{E}[\mathbf{A}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{A}(\mathbf{X} - \boldsymbol{\mu}))^\top] \\ &= \mathbb{E}[\mathbf{A}(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{A}^\top] \\ &= \mathbf{A}\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \mathbf{A}^\top \\ &= \boxed{\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top}.\end{aligned}$$

The translation vector $\mathbf{b}$ disappears because covariance depends only on deviations from the mean.

(c) For

$$Z = \mathbf{a}^\top \mathbf{X},$$

take $\mathbf{A} = \mathbf{a}^\top$ and $\mathbf{b} = 0$. Then

$$\boxed{\text{Var}(Z) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}}.$$

Also,

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \mathbb{E}\left[\left(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu})\right)^2\right] \geq 0.$$

Hence covariance matrices are positive semidefinite and

$$\boxed{\text{Var}(Z) \geq 0}.$$

Solution to Question 6: Bias–Variance Trade-off between Two Bernoulli Estimators

Since

$$S = \sum_{i=1}^n X_i \sim \text{Binomial}(n, p),$$

we have

$$\mathbb{E}[S] = np, \quad \text{Var}(S) = np(1-p).$$

Write $q = 1 - p$ when convenient.

(a) For

$$\begin{aligned} \hat{p}_1 &= \frac{S}{n}, \\ \mathbb{E}[\hat{p}_1] &= \frac{np}{n} = p, \end{aligned}$$

so

$$\boxed{\text{Bias}(\hat{p}_1) = 0}.$$

Also,

$$\boxed{\text{Var}(\hat{p}_1) = \frac{p(1-p)}{n}}.$$

For

$$\begin{aligned} \hat{p}_2 &= \frac{S+1}{n+2}, \\ \mathbb{E}[\hat{p}_2] &= \frac{np+1}{n+2}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Bias}(\hat{p}_2) &= \frac{np+1}{n+2} - p \\ &= \frac{1-2p}{n+2}. \end{aligned}$$

Hence

$$\boxed{\text{Bias}(\hat{p}_2) = \frac{1-2p}{n+2}}.$$

Adding a constant does not affect variance, so

$$\boxed{\text{Var}(\hat{p}_2) = \frac{np(1-p)}{(n+2)^2}}.$$

(b) Using

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2,$$

we obtain

$$\boxed{\text{MSE}(\hat{p}_1) = \frac{p(1-p)}{n}}.$$

For $\hat{p}_2$,

$$\boxed{\text{MSE}(\hat{p}_2) = \frac{np(1-p) + (1-2p)^2}{(n+2)^2}}.$$

(c) We require

$$\text{MSE}(\hat{p}_2) < \text{MSE}(\hat{p}_1).$$

Thus

$$\frac{np(1-p) + (1-2p)^2}{(n+2)^2} < \frac{p(1-p)}{n}.$$

Multiplying by the positive quantity $n(n+2)^2$ gives

$$n^2p(1-p) + n(1-2p)^2 < p(1-p)(n+2)^2.$$

Cancelling $n^2p(1-p)$ from both sides gives

$$\boxed{n(1-2p)^2 < 4(n+1)p(1-p)}.$$

Equivalently,

$$\boxed{p(1-p) > \frac{n}{4(2n+1)}}.$$

If desired, this may be written as

$$\boxed{\frac{1 - \sqrt{\frac{n+1}{2n+1}}}{2} < p < \frac{1 + \sqrt{\frac{n+1}{2n+1}}}{2}}.$$

Thus $\hat{p}_2$ can have smaller MSE for values of p sufficiently far from the boundaries 0 and 1. The estimator introduces bias, but its variance is smaller; MSE depends on the combined effect of variance and squared bias, not on bias alone.

Solution to Question 7: Sampling Distribution Induced by the Bootstrap

Condition on the observed dataset

$$D = \{x_1, \dots, x_n\}.$$

Each bootstrap draw X_j^* takes value x_i with probability $1/n$.

(a) The conditional bootstrap mean is

$$\begin{aligned}\mathbb{E}_*[X_j^*] &= \sum_{i=1}^n x_i \frac{1}{n} \\ &= \boxed{\bar{x}}.\end{aligned}$$

The conditional second moment is

$$\mathbb{E}_*[(X_j^*)^2] = \frac{1}{n} \sum_{i=1}^n x_i^2.$$

Therefore,

$$\begin{aligned}\text{Var}_*(X_j^*) &= \mathbb{E}_*[(X_j^*)^2] - (\mathbb{E}_*[X_j^*])^2 \\ &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \\ &= \boxed{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}.\end{aligned}$$

Define

$$s_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then

$$\boxed{\mathbb{E}_*[X_j^*] = \bar{x}, \quad \text{Var}_*(X_j^*) = s_n^2}.$$

(b) Because bootstrap draws are independent conditional on D ,

$$\begin{aligned}\mathbb{E}_*[\bar{X}^*] &= \mathbb{E}_*\left[\frac{1}{n} \sum_{j=1}^n X_j^*\right] \\ &= \frac{1}{n} \sum_{j=1}^n \mathbb{E}_*[X_j^*] \\ &= \boxed{\bar{x}}.\end{aligned}$$

Similarly,

$$\begin{aligned}\text{Var}_*(\bar{X}^*) &= \text{Var}_*\left(\frac{1}{n} \sum_{j=1}^n X_j^*\right) \\ &= \frac{1}{n^2} \sum_{j=1}^n \text{Var}_*(X_j^*) \\ &= \frac{1}{n^2} (n s_n^2) \\ &= \boxed{\frac{s_n^2}{n}}.\end{aligned}$$

The ideal bootstrap bias estimate for the sample mean is therefore

$$\mathbb{E}_*[\bar{X}^*] - \bar{x} = 0.$$

The corresponding bootstrap standard error is

$$\text{SE}_*(\bar{X}^*) = \sqrt{\frac{s_n^2}{n}}.$$

(c) Let the usual unbiased sample variance be

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Then

$$s_n^2 = \frac{n-1}{n} s^2.$$

Hence

$$\text{Var}_*(\bar{X}^*) = \frac{n-1}{n^2} s^2 = \frac{n-1}{n} \left(\frac{s^2}{n} \right).$$

Thus, relative to the familiar variance estimate s^2/n , the finite-sample multiplicative factor is

$$\frac{n-1}{n}.$$

For the standard error itself, the corresponding factor is

$$\sqrt{\frac{n-1}{n}}.$$

Solution to Question 8: Projection Geometry and OLS Variability

The OLS fitted vector is

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

(a) Because $\mathbf{X}^\top \mathbf{X}$ is symmetric and invertible,

$$\left[(\mathbf{X}^\top \mathbf{X})^{-1}\right]^\top = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Therefore,

$$\begin{aligned} \mathbf{H}^\top &= \left[\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top\right]^\top \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{H}. \end{aligned}$$

Hence $\mathbf{H}$ is symmetric.

Also,

$$\begin{aligned} \mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{H}. \end{aligned}$$

Thus

$$\mathbf{H}^\top = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{H}.$$

(b) The residual vector is

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}},$$

where

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Hence

$$\begin{aligned} \mathbf{X}^\top \mathbf{e} &= \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} \\ &= \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{0}. \end{aligned}$$

Therefore

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0},$$

so the residual vector is orthogonal to every column of $\mathbf{X}$.

For the leverage sum,

$$\sum_{i=1}^n h_{ii} = \text{tr}(\mathbf{H}).$$

Using cyclic invariance of the trace,

$$\begin{aligned} \text{tr}(\mathbf{H}) &= \text{tr} \left[\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \right] \\ &= \text{tr} \left[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \right] \\ &= \text{tr}(\mathbf{I}_p) \\ &= p. \end{aligned}$$

Thus

$$\sum_{i=1}^n h_{ii} = p.$$

(c) Conditioning on $\mathbf{X}$,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Because

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y},$$

its random part is $\mathbf{H}\boldsymbol{\varepsilon}$. Therefore

$$\begin{aligned} \text{Var}(\hat{\mathbf{y}} \mid \mathbf{X}) &= \mathbf{H} \text{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) \mathbf{H}^\top \\ &= \sigma^2 \mathbf{H} \mathbf{H}^\top \\ &= \sigma^2 \mathbf{H}^2 \\ &= \boxed{\sigma^2 \mathbf{H}}. \end{aligned}$$

For the residual vector,

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}.$$

Since

$$(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0},$$

we have

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}.$$

Therefore,

$$\begin{aligned} \text{Var}(\mathbf{e} \mid \mathbf{X}) &= (\mathbf{I} - \mathbf{H})\sigma^2 \mathbf{I}(\mathbf{I} - \mathbf{H})^\top \\ &= \sigma^2 (\mathbf{I} - \mathbf{H})^2 \\ &= \boxed{\sigma^2 (\mathbf{I} - \mathbf{H})}, \end{aligned}$$

because $\mathbf{I} - \mathbf{H}$ is also symmetric and idempotent.

(d) Taking diagonal elements gives

$$\text{Var}(\hat{y}_i \mid \mathbf{X}) = \sigma^2 h_{ii}$$

and

$$\text{Var}(e_i \mid \mathbf{X}) = \sigma^2 (1 - h_{ii}).$$

Thus higher leverage increases the conditional variability of the fitted value but decreases the conditional variability of the raw residual.

This matters because residuals at different observations do not have the same variance. A raw residual of a fixed magnitude is more unusual at a high-leverage point than at a low-leverage point, because its reference variance is smaller.

A variance-adjusted residual therefore uses the factor

$$\sqrt{1 - h_{ii}},$$

for example,

$$\frac{e_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}.$$

High-leverage observations can also pull the fitted model toward themselves, producing a small raw residual while still being influential. Therefore residual size and leverage should be interpreted together.